

Context

African Swine Fever (ASF) is a viral disease affecting multiple species of suids, such as wild boars (*Sus scrofa*) and domestic pigs (*Sus scrofa domesticus*).

It originates from Subsaharian Africa. After sporadical occurrences in other continents, the disease is now widespread (see timeline of the current epizootic).

It represents a threat for food production and biodiversity conservation.

20+

infected countries outside Africa, including China, Russia, Haiti, Germany, Italy, Spain, Thailand, Hungary...

21,500+

outbreaks reported to the World Organisation for Animal Health.

2007

The disease is reintroduced in Europe (Georgia)

2014

It reaches the European Union (Lithuania and Poland)

2018

The disease reaches eastern Asia (China)

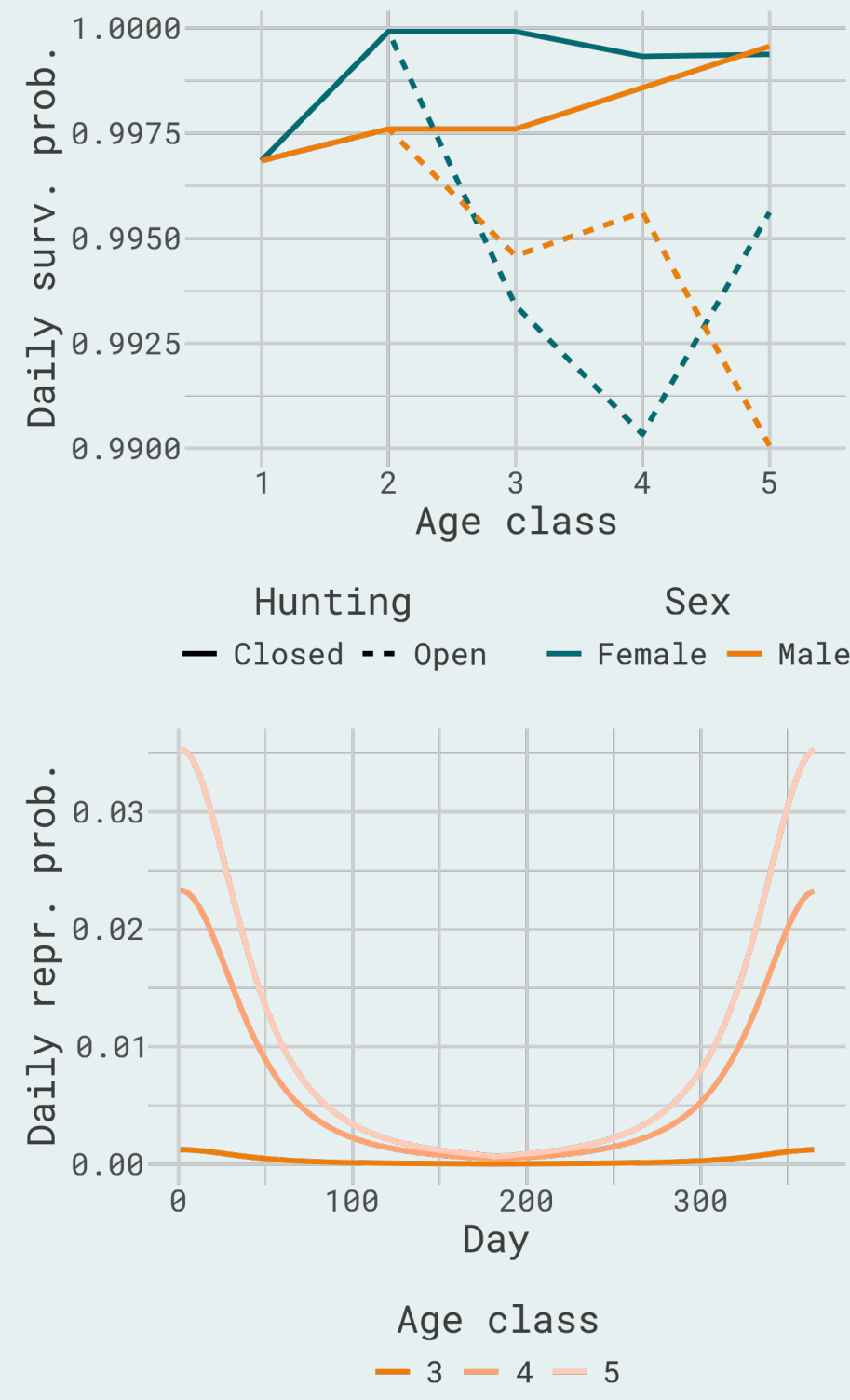
2021

The disease reaches the Americas (Dominican Republic and Haiti)

Methods

Bio-demographic component

→ Individuals reproduce, live and die. Parameters values are estimated from an eastern French population.



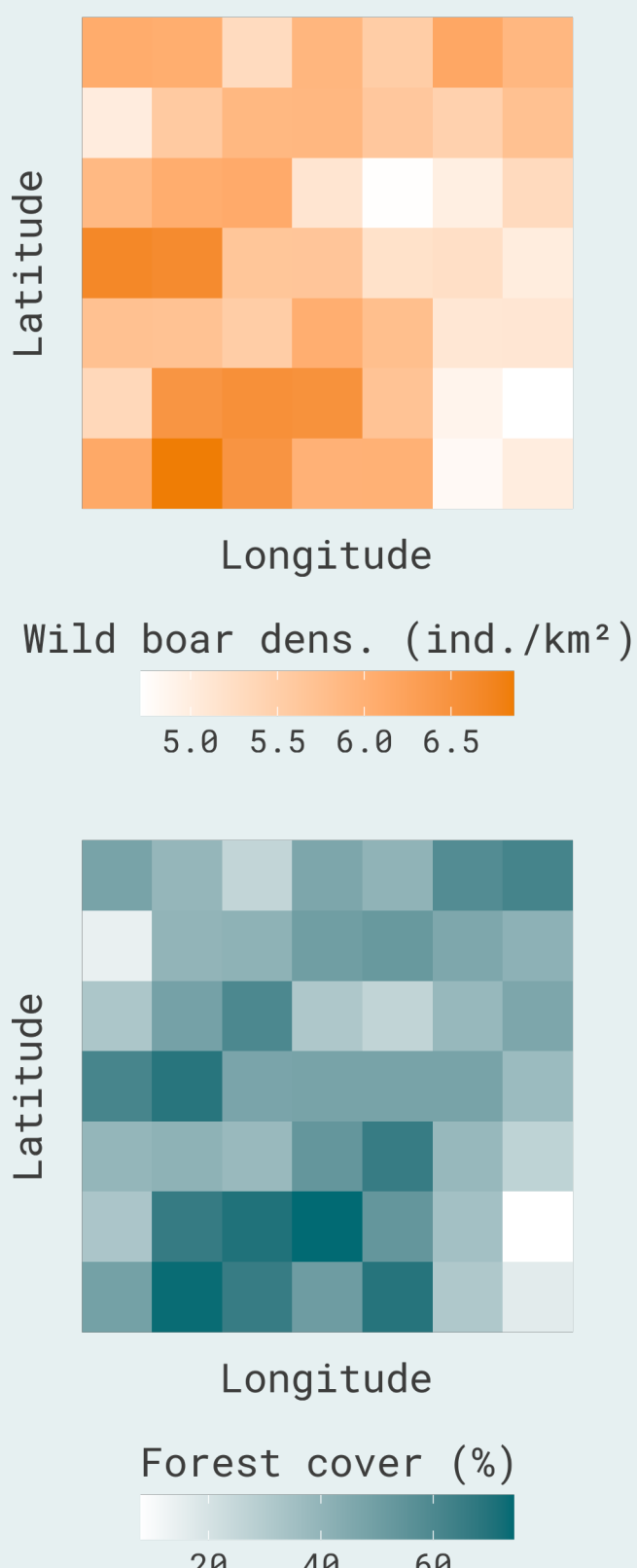
→ Survival based on age, sex and seasonality (Gamelon et al. 2012, 2021; Veylit et al. 2021).

→ Reproduction based on age and seasonality (Gamelon et al. 2011, 2012).

+ other parameters...

Spatial component

→ The population is divided in discrete spatial cells.



→ Each cell its initial population density (ENETWILD Consortium 2024), which changes with time (births, deaths, movements...).

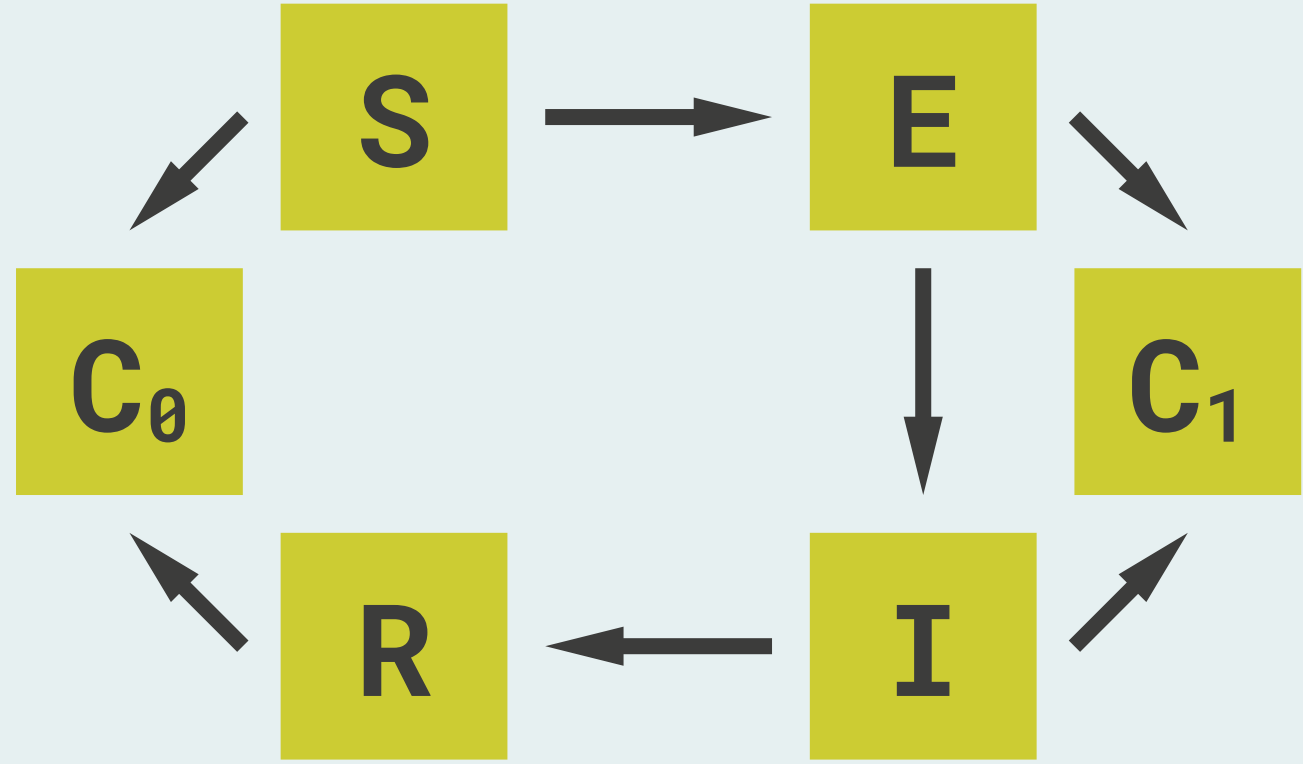
→ Individuals can move to neighbouring patches, with a preference for a higher forest cover (European Environment Agency, 2019).

Epidemiological component

→ SEIR stochastic individual-based model adapted from Fay et al. (2025).

→ Includes carcasses.

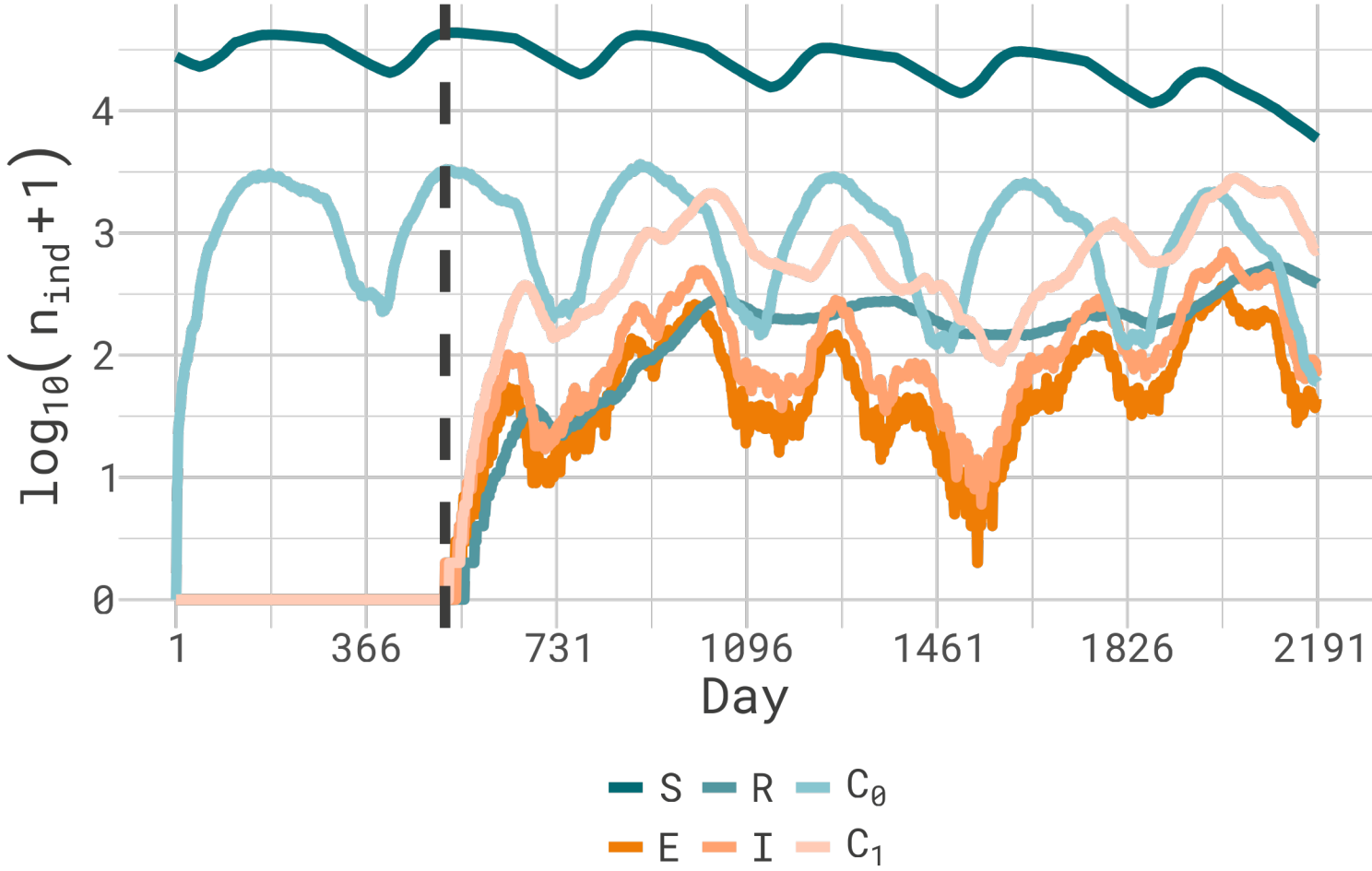
→ Different sources of infections: alive individuals, carcasses and environment.



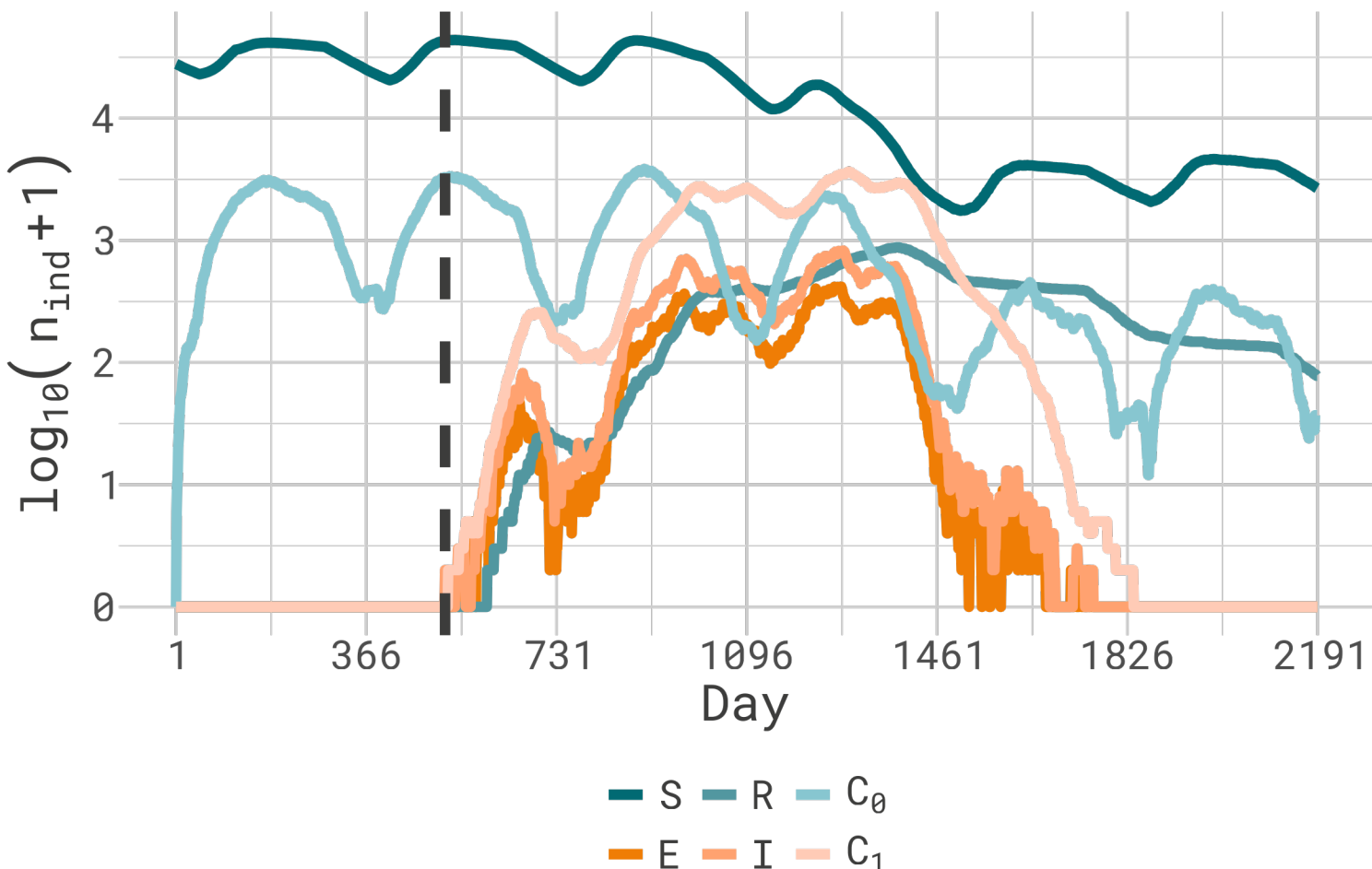
Model coded in the R language and simulations run on a computer cluster.

Results & Discussion

Exemple of simulation where endemicity is reached¹:



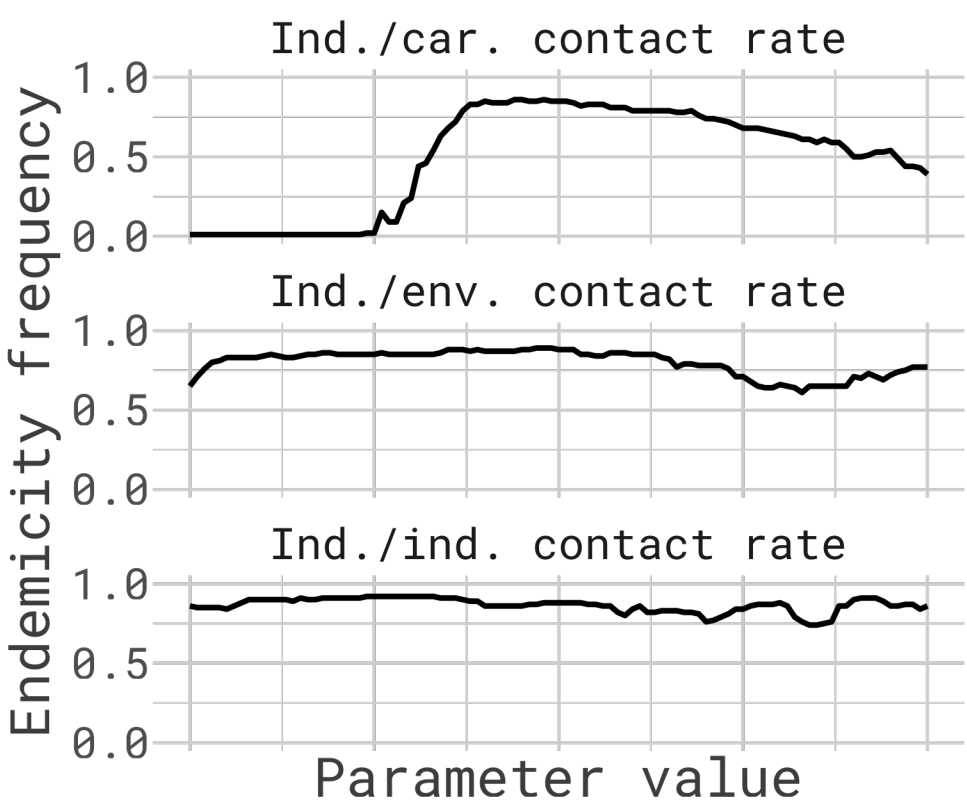
Exemple of simulation where endemicity is not reached:



Effect² of the different contact sources

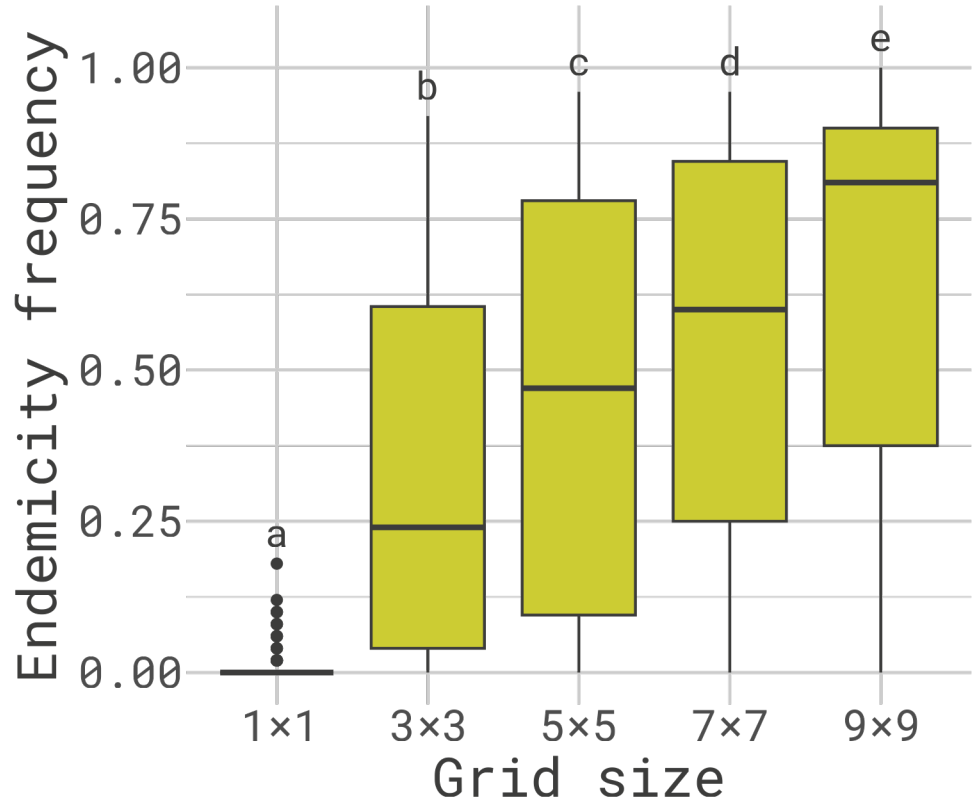
The risk of endemicity is mainly driven by the contacts with carcasses rather the contacts with other individuals or the infected environment.

This implies that carcass removal may be an efficient control strategy.



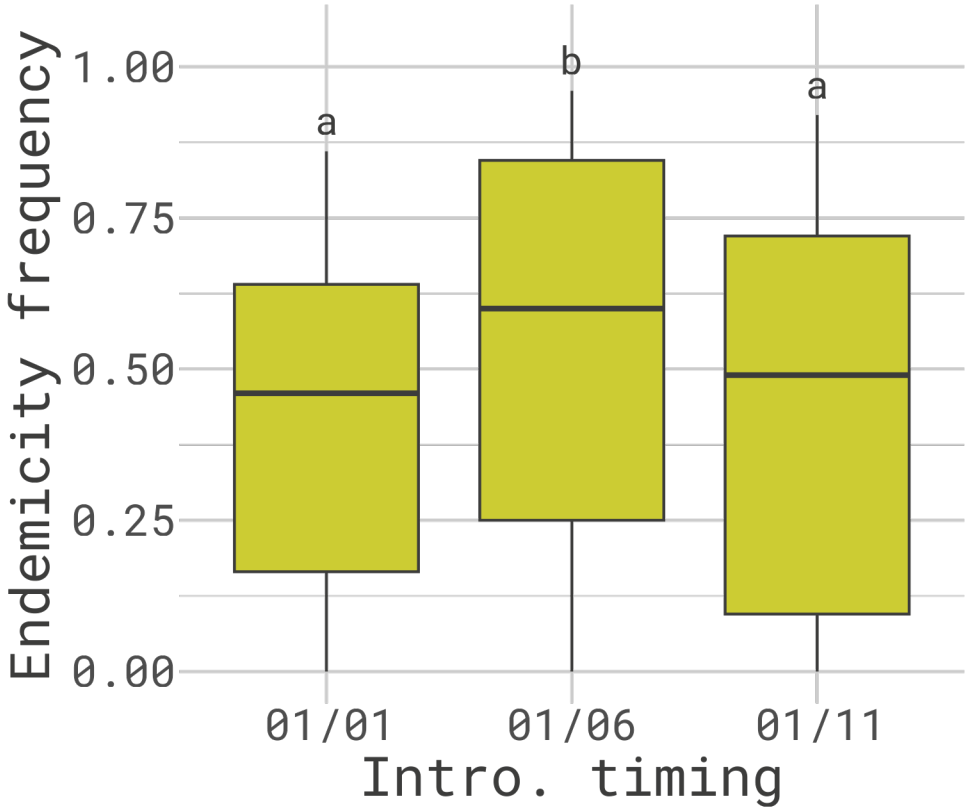
Effect³ of the spatial structure of the population

The largest the area in which the disease can spread, the higher the risk of endemicity. This implies that reducing the areas in wich wild boars can move (e.g., with fences) may be an efficient control method to prevent endemicity.



Effect³ of the introduction timing

Endemicity is observed more often if the virus is introduced in June rather than January or November. There is a probable link with the seasonal degradation rate of carcasses and the seasonal changes in the density of hosts.



Effect³ of the population density

Increasing (or decreasing) the population density decreases the risk of endemicity. This implies that using depopulation strategies may alternatively increase or decrease the risk of endemicity.

